

Turbo-Coded Optical Direct-Detection CDMA System with PPM Modulation

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Abstract—The performance of an optical code division multiple access (CDMA) system with turbo coding is analyzed and simulated. Turbo codes are parallel concatenated convolutional codes (PCCCs) in which the information bits are first encoded by a recursive systematic convolutional code, and then, after passing through an interleaver, are encoded by a second systematic convolutional encoder. Turbo coding is superimposed on an intensity-modulated optical channel with pulse-position modulation (PPM) and direct detection of the received optical signal, and the performance is evaluated in terms of an upper bound on bit error probability. From the simulation results, it is seen that turbo coding offers considerable coding gain over other methods, with reasonable encoding/decoding complexity. Also, it is demonstrated that, for a fixed code rate, the performance of the optical CDMA system can be substantially improved by increasing the interleaver length and the number of iterations. Serially concatenated convolutional codes (SCCCs) are also considered, and are shown to perform comparably to PCCC in general and better than PCCCs for the case of large signal photocounts. The results in this paper can be applied, for example, to indoor optical wireless LANs.

Index Terms—Optical code division multiple access (CDMA), pulse-position modulation (PPM), turbo codes.

I. INTRODUCTION

CODE DIVISION MULTIPLE ACCESS (CDMA) based on spread-spectrum technology is one of the most promising multiple access techniques for emerging generations of optical systems, whereby multiple users can be multiplexed at the same wavelengths and time slots through unique signature codes. The optical medium is very well suited to the CDMA technique in that it can provide extremely large bandwidths. By applying CDMA to an optical communication system, the following benefits can be obtained [1]–[8]: asynchronous access capability, accurate time of arrival measurements, flexibility of user allocation, ability to support variable bit rate and bursty traffic, security against unauthorized users, etc. Due to these advantages, CDMA has attracted much attention in the last two decades in both the radio-frequency and optical domains. From a practical point of view, optical CDMA systems are interesting since they require minimal optical signal processing and are virtually delay free, while from the theoretical point of view,

their performance depends on the auto- and cross-correlation properties of the sequences employed [5].

As a modulation format for optical communications, ON–OFF-keying (OOK) and (pulse-position modulation (PPM) have primarily been employed [7], [8]. For power-limited optical channels such as wireless optical channels, PPM is preferable to OOK because of its power efficiency. In PPM, a laser pulse is shifted into one of a set of possible pulse locations for data transmission. Each pulse position corresponds to a different PPM symbol.

There have been many approaches to improving the performance of optical CDMA systems such as multiuser detection, channel coding, etc. [2], [9]–[11]. Recently, in the channel coding community, there has been much interest in turbo codes [also termed parallel concatenated convolutional codes (PCCCs)] introduced by Berrou *et al.* in 1993 [9]. The substantial coding gain of turbo coding has been confirmed for CDMA systems in wireless channels as well as in AWGN channels [10]. The performance of optical communication systems with turbo coding has been examined in several recent works [12]–[14].

Channel coding schemes for optical communication systems have been employed to compensate for nonideal channel and receiver conditions in impaired photon channels such as deep-space, intersatellite, or wireless infrared channels [15]. In previous research, as a channel coding strategy for optical communications, Reed–Solomon (RS) coding has primarily been considered because the alphabet size can easily be matched to the PPM order, generating one-to-one correspondence between coded symbols and PPM symbols [17]. A limitation of RS codes is that they cannot use soft detector outputs directly. In contrast, turbo codes can make full use of soft information via the iterative decoding process.

Turbo decoding is performed in an iterative manner [16]. The encoder is formed by concatenation of two constituent codes in parallel and by separating the codes with an interleaver [17], [18]. This interleaver permutes the time order of the information sequence to be transmitted and then uses the permuted sequence as an input to the second component encoder. It is known that turbo coding performance is enhanced with increasing interleaver length, and decoding complexity grows linearly, making the decoding of large block lengths possible [16]. Turbo decoding is not maximum likelihood (ML) decoding, but tries to approach ML decoding in an iterative way. At every iteration, a single decoding is performed using the channel observations as well as reliability information delivered by other decoders that have acted before.

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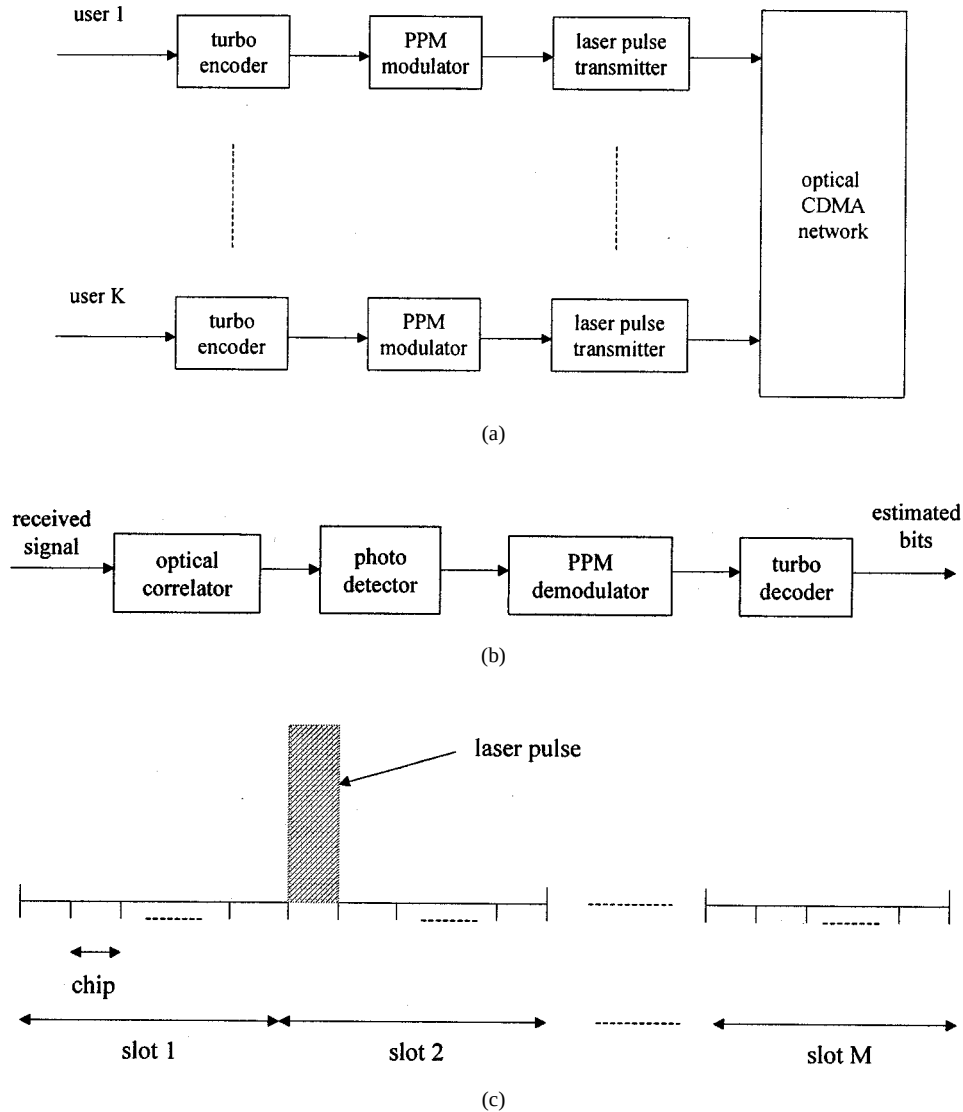


Fig. 1. Block diagram of an optical CDMA system. (a) Transmitter. (b) Receiver. (c) PPM signaling format.

In this paper, the performance of an optical PPM/CDMA system with turbo coding is analyzed and simulated. It is assumed that the optical channel is an intensity-modulated (IM) channel with direct-detection of the received optical signal. The performance is evaluated in terms of an upper bound on bit error probability. For the turbo decoding, the maximum *a posteriori* probability (MAP) algorithm and its modified versions such as the Max-Log-MAP algorithm and the soft output Viterbi algorithm (SOVA) are considered. Serially concatenated convolutional codes (SCCCs) are also discussed, and are compared with the turbo codes.

This paper is organized as follows. In Section II, a system model of an optical CDMA system and structures for the turbo encoder and decoder are described. In Section III, a receiver for an optical CDMA system is modeled, and its corresponding bit error probability is derived. In Section IV, the turbo decoding algorithms are described, while in Section V encoding and decoding of SCCCs are discussed. Simulation results are presented in Section VI, and conclusions are drawn in Section VII.

II. SYSTEM MODEL

A. System Description

Fig. 1(a) and (b) shows block diagrams of a transmitter and a receiver for the optical PPM/CDMA system of interest. The information bit sequence of each user is fed into a turbo encoder. The turbo-encoded bit stream is then modulated in a PPM modulator where the coded bit stream is first blocked into a symbol of length $\log_2 M$ data bits. The transmitter then sends a pulse sequence in one of M time slots to represent the given block. The slot represents the duration of each symbol in the M -ary PPM signaling format which has M slots; i.e., the laser is pulsed on at the first chip position corresponding to the given symbol.

The optical CDMA network consists of K users, each transmitting its own message. Each user is assigned a unique optical pulse sequence for its address. The optical CDMA network has the advantages of being able to use simple pulsed lasers without wavelength control and standard wideband photo detectors without narrow optical filters. All of the users operate

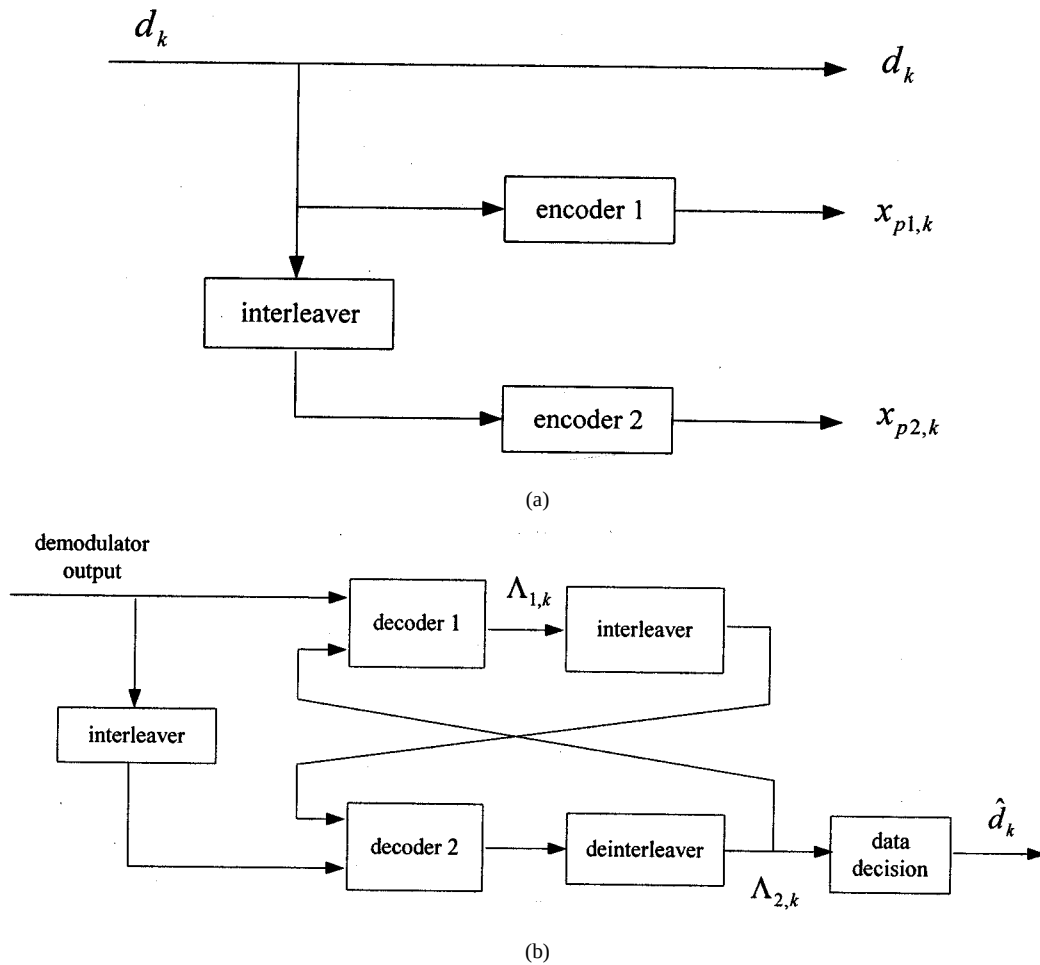


Fig. 2. Structure of the turbo code—parallel concatenated convolutional code (PCCC). (a) Encoder. (b) Decoder.

independently, and no common clock is needed to synchronize the transmitters of the users. At the physical level, this network is typically a passive optical network with a star topology, while at the logical level, this network is a type of broadcast-and-select network.

The received signal is first processed in the optical correlator, which is a set of optical delay lines matched to the pulse positions. The optical signals of the users are additive in intensity because the laser lights are assumed to be noncoherent. When the desired signal passes through the optical correlator, the correlator output is converted into an electrical signal at the photodetector. Then, in the PPM demodulator, the position giving the highest output is selected as containing the transmitted symbol. After this detection process, the output of the PPM demodulator is decoded by the turbo decoder to estimate the transmitted information bits.

Fig. 1(c) depicts the M -ary PPM signaling scheme. The laser is pulsed on at the first chip of the slot that represents the transmitted symbol. In this case, the laser pulse is placed in the second slot as an example. The optical correlator at the receiver is shown in Fig. 1(d) [19], [20]. An individual receiver observes the sum of all the pulse transmissions from K transmitters, and recovers the data from a particular transmitter by using a matched pulse code correlator. This correlator can be accomplished optically by the use of parallel fiber delay lines, and the

delays are matched to the desired sequence. If the sequence is matched to the desired transmitter's sequence, this correlator produces high correlator output. The photodetector following the optical correlator detects the optical correlation peak. The received sequences from other transmitters which do not match the correlator delays will correlate to lower values, which produce cross correlation crosstalk.

B. Turbo Encoding/Decoding

Fig. 2(a) and (b) shows block diagrams of the turbo encoder and decoder. The turbo encoder consists of two recursive systematic convolutional codes known as *constituent codes*. Since the turbo code is a type of linear block code, the encoding operation can be viewed as the modulo-2 matrix multiplication of an information matrix with a generator matrix. Encoder 1 encodes the input data sequence directly, while Encoder 2 encodes the data sequence permuted in time by a pseudorandom interleaver with length N . The encoder outputs are composed of the systematic bit d_k and parity bits $x_{p1,k}$ and $x_{p2,k}$ from the two constituent codes. In this encoding process, a very large effective constraint length is generated through interleaving and concatenation. For this reason, the conventional Viterbi decoding algorithm is not feasible for turbo decoding, and turbo decoding is typically performed in an iterative manner. In particular, each

constituent code is separately decoded using the most recent decoding information from the other constituent code. Each constituent decoder computes the *a posteriori* log-likelihood ratio (LLR) of the systematic bits, which is given by

$$\Lambda_k = \log \frac{p(d_k = 1|y_1, y_2, \dots, y_N)}{p(d_k = 0|y_1, y_2, \dots, y_N)} \quad (1)$$

where $y_i (1 \leq i \leq N)$ is the decoder output and $p(d_k = j|y_1, y_2, \dots, y_N)$ is the *a posteriori* probability of the bit value j . This extrinsic information Λ_k can be regarded as a type of diversity in that it can refine the decoder outputs in each iteration. The decoding process continues until some stopping criterion is met. After iteration, the data bit decision $\hat{d}_k$ is made based on the final decoder output.

The turbo code integrates code concatenation in a pseudo-random approach where the randomness and long block size are provided by the interleaver. The first decoder passes the extrinsic information [a part of the soft output provided by an *a posteriori* probability (APP)] to the next decoding stage. The term “turbo” stems from the fact that the decoder uses its processed output as *a priori* input in the next iteration, and thus reuses the information in the observations.

III. PERFORMANCE ANALYSIS

To analyze the performance of this system, we make the following assumptions:

- 1) the optical characteristics of all K users are identical,
- 2) synchronization between the transmitter and the intended receiver is perfect,
- 3) the constituent codes of the turbo encoder are identical, and
- 4) the signal photon rate is constant for all the transmitters.

A. Optical Orthogonal Codes

Optical CDMA systems with direct detection require signature sequences composed of 0s and 1s with desirable correlation properties, since this type of system has basically unipolar (positive) properties. To obtain the smallest multiple access interference (MAI), codes with the smallest cross correlations are necessary. Moreover, small out-of-phase autocorrelations are also required for acquiring and maintaining synchronization between the transmitter and the desired receiver. In this paper, it is assumed that a low-weight optical orthogonal code (OOC) with length F and weight w is assigned to each user as a signature code. The OOC of the k th user is $c^k = (c_0^k, c_1^k, \dots, c_{F-1}^k)$, $c_i^k \in \{0, 1\}$, where the number of elements of c^k with $c_i^k = 1$, ($0 \leq i \leq F-1$) is equal to w . Then, the correlation between codes is given by

$$R_{c^k c^l}(j) = \sum_{i=0}^{F-1} c_i^k c_{i-j}^l, \quad (2a)$$

$$= \begin{cases} w, & \text{for } k=l \text{ and } j=0, \\ \leq 1, & \text{for } k=l \text{ and } 1 \leq j \leq F-1, \\ \leq 1, & \text{for } k \neq l \text{ and } 0 \leq j \leq F-1, \end{cases} \quad (2b)$$

where c_{i-j}^l is a j -times left rotation code of c_i^l for integers (modulo F) i and j . In this weight-one coincidence OOC, maximum cross correlation and maximum off-peak autocorrelation are bounded by unity for all k, l , and j . For OOCs with length F and weight w , the maximum admissible number of codes is bounded by [6]

$$K \leq \left\lfloor \frac{F-1}{w(w-1)} \right\rfloor \quad (3)$$

where K is the number of users and $\lfloor x \rfloor$ denoted the integer part of x .

B. Receiver Modeling

For M -ary PPM signaling, each input symbol is encoded into one of M slot positions with slot duration T_s . Each slot consists of F chips where F is the processing gain (or spreading factor) for the CDMA signals. The OOC for the k th user can be written as

$$c^k(t) = \sum_{i=-\infty}^{\infty} c_i^k P_{T_c}(t - iT_c) \quad (4)$$

where $\{c_i^k\}$ is interpreted as a periodic sequence with period F , $T_c = T_s/F$ is the chip duration, and $P_{T_c}(\cdot)$ is unit-amplitude rectangular pulse of duration T_c ; i.e.,

$$P_{T_c}(t) = \begin{cases} 1, & \text{for } 0 < t < T_c, \\ 0, & \text{for } \text{otherwise.} \end{cases} \quad (5)$$

The output rate of the laser signal for the k th user is given by

$$b^k(t) = \sum_{j=-\infty}^{\infty} \lambda_s w^2 P_{T_c}(t - b_j^k T_s - jT) \quad (6)$$

where

- $\lambda_s w^2$ photon absorption rate of the signal before the optical correlator;
- λ_s signal photon rate of each tapped optical delay line;
- b_j^k determines one of the M slots;
- $T = MT_s$ PPM symbol duration.

The output of the laser is converted into a specific train of pulses according to the assigned OOC, where the photon rate of each pulsed chip is $\lambda_s w$. Thus, the received signal is given by

$$r(t) = \sum_{k=1}^K \frac{1}{w} b^k(t - \tau_k) c^k(t - \tau_k) + n(t) \quad (7)$$

where τ_k is the delay associated with the k th user and $n(t)$ is noise due to dark current. In this paper, a chip synchronous system is assumed because it offers an upper bound on bit error probability for the general case. Thus, the delays $\tau_k (1 \leq k \leq K)$ are assumed to be integer multiples of T_c .

We assume that detection is implemented with an avalanche photodiode (APD), in which case the average number of photons absorbed by the detector over a chip duration is given by

$$n_p = \frac{\eta}{h\nu} \int_0^{T_c} P(t) dt \quad (8)$$

where

- $P(t)$ received optical power;
- h Plank's constant ($=6.624 \times 10^{-34}$);
- ν optical frequency;
- η quantum efficiency of the APD.

For constant received optical power, $P(t) = P$, the average number of absorbed photons becomes $n_p = \eta PT_c/h\nu = \lambda_s T_c$ where $\lambda_s = \eta P/h\nu$ is the photon absorption rate. The probability that n photons are absorbed by the detector over a chip duration is given by the Poisson distribution with arrival rate n_p .

When the gain of the detector is high enough, the effect of thermal noise can be neglected, and the receiver is shot-noise limited. In this case, the output in each slot is directly proportional to the photocount over that slot. Thus, a sufficient statistic for detection is the number of photons received in each of the observation intervals. These photocounts can be modeled as conditional Poisson random variables. Assuming, without loss of generality, that the user of interest is user 1, then the APD output in the i th slot can be written as

$$Z_i = D_i + \sum_{k=2}^K I_i^k + N_i, \quad 1 \leq i \leq M \quad (9)$$

where D_i , I_i^k , and N_i are Poisson photocount components due to the desired signal (i.e., user 1's signal), user k 's signal, and noise, respectively. (In a direct-detection optical CDMA system, the optical noise component is additive to the optical signal component in intensity.) The mean value of N_i is given by

$$E[N_i] = w\lambda_b T_c \quad (10)$$

where λ_b is the photon absorption rate of the detector due to background noise or dark current. The mean value of I_i^k conditioned on the first bit of the k th user is given by

$$E[I_i^k | b_1^k] = R_{c^1 c^k} \lambda_s T_c \delta_{j_k i} \quad (11)$$

where $j_k = b_1^k$, $\delta_{j_k i}$ denotes the Kronecker delta function, and $R_{c^1 c^k}$ is the cross correlation between the first and k th users. Assuming the worst case cross correlation ($R_{c^1 c^k} = 1$), the mean photocount due to the $K - 1$ interfering users is given by

$$E \left[\sum_{k=2}^K I_i^k \middle| \{b_1^2, b_1^3, \dots, b_1^K\} \right] = \lambda_s T_c \kappa_i \quad (12)$$

where κ_i denotes the number of interfering users hitting the i th slot of user 1. The parameter κ_i is a binomial random variable, i.e.,

$$\Pr\{\kappa_i = j\} = \binom{K-1}{j} \left(\frac{1}{M}\right)^j \left(1 - \frac{1}{M}\right)^{K-1-j} \quad j = 1, 2, \dots, K. \quad (13)$$

C. Uncoded Optical CDMA System

The word error probability is given by

$$P_{w,u} = 1 - P_{c,u} \quad (14)$$

where $P_{c,u}$ is the probability of making a correct slot decision. The slot decision rules are as follows: 1) symbol i is chosen if $Z_i > Z_j$ for all $j \neq i$ and 2) when the photocounts between Z_i and Z_j are the same, any one of the two corresponding symbols is randomly chosen. Under the assumption of equally likely data symbols, the uncoded probability of correct slot decision is lower bounded by

$$P_{c,u} \geq \sum_{i=1}^M \Pr\{Z_i > Z_1, \dots, Z_i > Z_{i-1}, Z_i > Z_{i+1}, \dots, Z_i > Z_M | b_1 = i\} \cdot \Pr\{b_1 = i\} \quad (15a)$$

$$= \Pr\{Z_1 > Z_2, Z_1 > Z_3, \dots, Z_1 > Z_M | b_1 = 1\} \quad (15b)$$

$$= \sum_{l_1=0}^{K-1} \sum_{l_2=0}^{K-1-l_1} \dots \sum_{l_M=0}^{K-1-l_s} P_{c1} \cdot \Pr\{\kappa = 1\} \quad (15c)$$

where $P_{c1} = \Pr\{Z_1 > Z_2, Z_1 > Z_3, \dots, Z_1 > Z_M | b_1 = 1, \kappa = 1\}$, $1 = \{l_1, l_2, \dots, l_M\}$, $l_s = \sum_{i=1}^M l_i$, and $\kappa = \{\kappa_1, \kappa_2, \dots, \kappa_M\}$. Note that κ has a multinomial distribution given by

$$\Pr\{\kappa = 1\} = \frac{1}{M^{K-1}} \frac{(K-1)!}{l_1! l_2! \dots l_M!}. \quad (16)$$

From the Poisson distribution of the APD output, P_{c1} is obtained as

$$P_{c1} = \sum_{k=1}^{\infty} POS(k, K_s + K_b + \lambda_s T_c l_1) \cdot \prod_{j=2}^M \left[\sum_{i=0}^{k-1} POS(i, K_b + \lambda_s T_c l_j) \right] \quad (17)$$

where $POS(x, y) = y^x e^{-y} / x!$ denotes the Poisson mass function, and where $K_s = w\lambda_s T_c$ and $K_b = w\lambda_b T_c$ are average photocounts per symbol due to the desired signal and noise, respectively. Finally, the uncoded bit error probability is bounded by

$$P_{b,u} \leq \frac{1}{2} \frac{M}{M-1} P_{w,u}. \quad (18)$$

D. Turbo-Coded Optical CDMA System

An upper bound on the turbo-coded bit error probability can be obtained from the weight distribution of the code. However, it is difficult to compute the weight distribution of a turbo code with a particular interleaver. Therefore, it is assumed that a uniformly drawn random interleaver is employed. In this case, the average of the weight distributions over all possible interleavers is taken, which corresponds to mapping an input sequence into all its distinct permutations with equal probability.

In a rate-1/3 turbo code as described above, the three bits at the encoder output can be interpreted as the sum of one uncoded bit and two parity bits. We typically refer to these rate-one outputs from each encoder as code fragments. Thus, the turbo code becomes a parallel concatenation of the code fragments. The

transfer function (or generator function) of the constituent encoder is given by [18]

$$G(X, I, D) = \sum_{x \geq 0} \sum_{i \geq 0} \sum_{d \geq 0} X^x I^i D^d g(x, i, d) \quad (19)$$

where $g(x, i, d)$ is the number of paths of length x , input Hamming weight i , and output Hamming weight d . The conditional probability of producing a codeword fragment of Hamming weight d given a randomly selected input sequence of Hamming weight i is given by

$$p_1(d|i) = \frac{g(N, i, d)}{\sum_d g(N, i, d)} \quad (20a)$$

$$= \frac{g(N, i, d)}{\binom{N}{i}} \quad (20b)$$

where N is the number of information bits and $\sum_d g(N, i, d)$ is the total number of codewords of Hamming weight i , given by $\binom{N}{i}$. For a rate-1/3 turbo encoder, $p_1(d|i)$ is modified into [18]

$$p_2(d_1, d_2|i) = \frac{g_1(N, i, d_1)g_2(N, i, d_2)}{\binom{N}{i}} \quad (21)$$

where $g_1(N, i, d_1)$ and $g_2(N, i, d_2)$ are the $g(N, i, d)$ values for the first and second encoders, respectively. The conditional transfer function for the code fragment with length N is given by

$$G(N, I, D) = \sum_{i=0}^N I^i \sum_{d_1=0}^N \sum_{d_2=0}^N p_2(d_1, d_2|i) D^{d_1} D^{d_2} \quad (22a)$$

$$= \sum_{i=0}^N I^i \sum_{d=0}^{2N} p_2(d|i) D^d \quad (22b)$$

where $d = d_1 + d_2$ and $d = \{d_1, d_2\}$. Then, the turbo-coded bit error probability is upper bounded by

$$P_b \leq \sum_{i=0}^N \sum_{d=0}^{2N} \frac{i}{N} A(i, d) P_{b,u} \quad (23)$$

where $P'_{b,u}$ is $P_{b,u}$ with P_{c1} replaced by P'_{c1} and $A(i, d)$ is the number of codewords generated by information sequence of weight i and a parity bit stream of weight d , and is given by

$$A(i, d) = \sum_{d_1=0}^N \sum_{d_2=0}^N p_2(d_1, d_2|i). \quad (24)$$

From (17), the term P'_{c1} is given by

$$P'_{c1} = \sum_{k=1}^{\infty} POS(k, (K_s + K_b + \lambda_s T_c l_1)(i + d)) \cdot \prod_{j=2}^M \left[\sum_{r=0}^{k-1} POS(r, (K_b + \lambda_s T_c l_j)(i + d)) \right] \quad (25)$$

where $i + d$ is the Hamming weight of the codeword, and comes from the fact that the desired path at the trellis and the all zero path differ in $i + d$ bit positions [18].

IV. TURBO DECODING ALGORITHM

Optimum ML decoding algorithms for turbo codes are very complex due to the presence of the interleaver. However, iterative decoding of such codes is quite feasible and often leads to near-capacity performance. Iterative decoding involves iteration between constituent decoders, with an exchange of soft information between iterations. Ideally, these constituent decoders should be MAP decoders, in order to compute posterior bit probabilities. However, since the MAP decoding algorithm is also very complex, alternative algorithms with lower complexity have been proposed for the constituent decoders. One of these is the SOVA, which is a modification of the standard Viterbi algorithm to produce soft outputs. For both the MAP and SOVA algorithms, the size of the encoder registers, and hence the constraint lengths of the component codes, is severely limited since the decoding complexity (measured by the number of trellis states) increases exponentially with constraint length. To make implementation easier, the Max-Log-MAP algorithm has also been proposed for this purpose.

A. MAP Algorithm

Each constituent convolutional encoder operates via a sequential process, in which a given information bit, say d_k , is associated with a transition from step $k - 1$ to step k in the process. The MAP decoding algorithm makes use of the branch transition metric corresponding to each such transition, which is given by [21], [22]

$$\begin{aligned} \gamma_i(A_k) &= \Pr(y_k^s | d_k = i) \cdot \Pr(y_k^p | d_k = i, S_k = m, S_{k-1} = m') \\ &\quad \cdot \Pr(d_k = i | S_k = m, S_{k-1} = m') \\ &\quad \cdot \Pr(S_k = m | S_{k-1} = m') \end{aligned} \quad (26)$$

where S_k is the state of the encoder at time k , $(m, m') \in X_k^i$ with X_k^i the set of transitions $S_{k-1} = m \rightarrow S_k = m'$ that are caused by $d_k = i$, and y_k^s and y_k^p are channel outputs for d_k and $x_{p,k}$, respectively, the latter being the parity bit associated with the particular constituent code being decoded. Here, A_k denotes the collection $(y_k^s, y_k^p, S_{k-1}, S_k)$. Through these branch metrics, the two channel outputs y_k^s and y_k^p are used to compute an LLR of *a posteriori* probabilities, as described below. This LLR is then used to produce *a priori* probabilities for the succeeding MAP decoder (i.e., the one for the other constituent code) after which a new LLR is evaluated, and used in turn by the first decoder to update its LLR. This procedure is repeated many times until convergence.

In the MAP algorithm, all possible paths in the trellis are considered to optimally determine the reliability of information data bits. MAP values can be computed through the BCJR algorithm [21], which has been shown empirically to perform remarkably well and to converge to the optimal decoding solution. The BCJR decoding algorithm differs from the Viterbi algorithm in two ways:

- 1) The BCJR algorithm produces soft outputs, while the Viterbi algorithm outputs either 0 or 1 for each estimated bit.
- 2) The BCJR algorithm attempts to minimize bit error probability by estimating the APP of the individual bits of the codeword, while the Viterbi algorithm tries to minimize codeword error probability by finding a ML estimate of the transmitted codeword.

The turbo decoding process can be regarded as the iterative improvement of the APP because the LLR is the ratio of the APPs of each information bit value: $d_k = 1$ and $d_k = 0$. The turbo decoder makes full use of the APP (soft information) generated by the previous MAP decoder as *a priori* information. The APP is given by [21], [22]

$$\Lambda(d_k) = \log \frac{\sum_{S_k} \sum_{S_{k-1}} \gamma_1(A_k) \cdot \alpha_{k-1}(S_{k-1}) \cdot \beta_k(S_k)}{\sum_{S_k} \sum_{S_{k-1}} \sum_{i=0}^1 \gamma_i(A_k) \cdot \alpha_{k-1}(S_{k-1}) \cdot \beta_k(S_k)} \quad (27)$$

where $\alpha_k(\cdot)$ and $\beta_k(\cdot)$ are forward and backward recursion parameters, respectively, given by [21], [22]

$$\alpha_k(S_k) = \frac{\sum_{S_k} \sum_{i=0}^1 \gamma_i(A_k) \cdot \alpha_{k-1}(S_{k-1})}{\sum_{S_k} \sum_{S_{k-1}} \sum_{i=0}^1 \gamma_i(A_k) \cdot \alpha_{k-1}(S_{k-1})} \quad (28)$$

and

$$\beta_k(S_k) = \frac{\sum_{S_{k+1}} \sum_{i=0}^1 \gamma_i(A_{k+1}) \cdot \beta_{k+1}(S_{k+1})}{\sum_{S_k} \sum_{S_{k+1}} \sum_{i=0}^1 \gamma_i(A_{k+1}) \cdot \beta_{k+1}(S_{k+1})}. \quad (29)$$

The evaluation of $\Lambda(d_k)$ thus consists of the following steps:

- Step 1) the initial conditions are $\alpha_0(S_0) = 0$ for $S_0 = 0$ and $\beta_N(S_N) = 1$ for $S_N = 0$;
- Step 2) compute $\gamma_i(A_k)$ from (26);
- Step 3) compute $\alpha_k(S_k)$ from (28);
- Step 4) compute $\beta_k(S_k)$ from (29) and the results of Step 2) and Step 3);
- Step 5) compute $\Lambda(d_k)$ from (27).

B. Max-Log-MAP Algorithm

Numerical stability problems can arise in the computations of $\gamma_i(\cdot)$, $\alpha(\cdot)$, and $\beta(\cdot)$ due to the large dynamic ranges of these quantities. This problem can be solved by performing computations in the logarithmic domain. Moreover, significant computational savings also arise in this situation if we make the approximation $\log(e^{x_1} + e^{x_2} + \dots + e^{x_n}) = \max\{x_1, x_2, \dots, x_n\}$ in the computations, and the resulting algorithm is called the Max-Log-MAP algorithm. Although this technique solves computational complexity and dynamic range issues, it can still require a large memory.

C. SOVA Algorithm

Decoder complexity can be reduced further by using the SOVA. The traditional Viterbi algorithm is a forward dynamic programming solution to the problem of ML decoding of convolutional codes. In the conventional Viterbi algorithm, the path with the largest metric is selected as a survivor at each node by maximizing over all the possible paths [23]. In the hard decision version, the Max-Log-MAP and SOVA algorithms make use of the same metric. However, in the soft decision version, these two algorithms operate differently. In the SOVA algorithm, two paths are considered, but the competing path may not be the best competing path. Usually, the competing path determines the reliability of soft information and survives to merge with the ML path. The SOVA algorithm computes only one competing path at every decoding step. Thus, for each information bit, it considers only one survivor of the Viterbi algorithm.

Like Max-Log-MAP, SOVA processing is also performed in the logarithmic domain, while executing the same operations as in the Viterbi algorithm. The SOVA path-update process generates soft information (reliability information about the bits along the survivor paths) as well as hard outputs. A fundamental difference between the MAP algorithm and the SOVA is that the MAP algorithm minimizes bit error probability while the SOVA minimizes sequence error probability.

V. SERIALY CONCATENATED CONVOLUTIONAL CODES (SCCCs)

A. Basic Principle of SCCC

Serially concatenated coding was proposed by Forney [24] to yield codes that have an exponentially decreasing probability of error with an algebraically increasing decoding complexity. The conventional concatenated code consists of the cascade of an inner code and an outer code. Typically, some kind of soft information is exchanged between the constituent codes, and the optimum output of the inner decoder is the form of an APP. In most applications, the inner code is a relatively short code (typically a convolutional code) admitting simple ML decoding, whereas the outer code is a long high-rate algebraic code (typically, a nonbinary RS code) equipped with a powerful algebraic error-correction algorithm. The concatenated code is appropriate for systems that require high coding gain such as deep-space communications.

The concatenation of two convolutional codes, either in a series or in parallel has been proposed [25], [26] as an alternative to conventional concatenated codes. As noted previously, turbo codes are parallel concatenated convolutional codes where the information bits are first encoded by a recursive systematic convolutional code, and then, after passing through an interleaver, are encoded by a second systematic convolutional encoder. The turbo-encoded sequences are formed by the information bits and parity bits generated by the two constituent encoders. An alternative approach to using two parallel convolutional encoders and an interleaver is the SCCC, which has been shown to yield performance comparable, and for some cases superior, to the PCCC [25], [26]. (One can also serially concatenate block codes [26], but the SCCC tends result in higher inter-

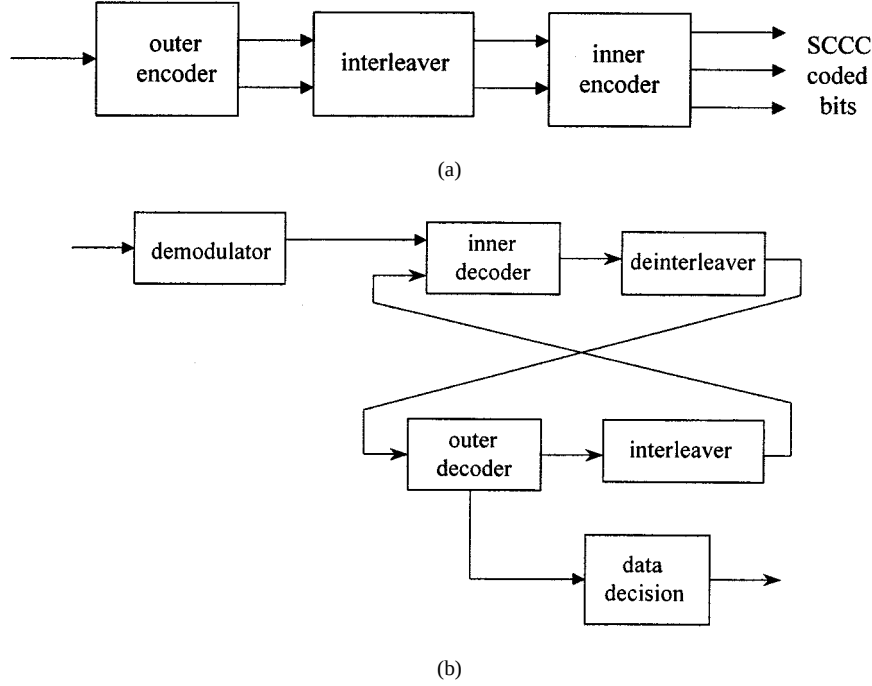


Fig. 3. Structure of the SCCC. (a) Encoder. (b) Decoder.

leaver gain and lower-complexity MAP decoding than serially concatenated block codes.)

B. Structure of SCCC

The block diagrams shown in Fig. 3(a) and (b) describe the encoder/decoder of the SCCC. The SCCC encoder shown in Fig. 3(a) consists of an inner encoder with code rate R_i and an outer encoder with code rate R_o . The overall code rate becomes $R_c = R_i R_o$. An interleaver permutes the time order of the output codewords of the outer encoder before passing them to the inner encoder. (The purpose of the interleaver is to separate bursts of errors produced by the inner decoder.) The SCCC decoder shown in Fig. 3(b) consists of an outer decoder, an inner decoder, and an interleaver and deinterleaver. The inner decoder employs a soft-output decoding algorithm to provide soft-input decisions as inputs to the outer decoder. The primary difference between the SCCC and PCCC decoder structures is that, in the SCCC decoder, the outer decoder does not make use of a direct input from the demodulator.

C. Comparison between SCCC and PCCC

The performance of PCCCs is known to be limited by the Hamming weight of error events of the code generated by information sequences of lowest weight w_m . For the PCCC, the decrease of bit error probability is proportional to N^{1-w_m} , which describes the *interleaver gain* [17]. An error event of a convolutional code is a sequence diverging from the zero state at time zero and merging again into the zero state after some time [17]. The decrease in bit error probability for the PCCC is at a rate of $1/N$ when the PCCC with two recursive constituent encoders has $w_m = 2$. In contrast, the performance limitation of the SCCC comes from error events of the inner code produced by its input sequences of lowest weight. Since the information

sequence for the inner encoder of the SCCC is constrained, the lowest weight is determined by the outer encoder. The interleaver gain of the SCCC is given by

$$I_g = \begin{cases} N^{-d_{f,o}/2}, & \text{for even } d_{f,o} \\ N^{-(d_{f,o}+1)/2}, & \text{for odd } d_{f,o} \end{cases} \quad (30)$$

where $d_{f,o}$ is the free distance of the outer code, which is typically larger than two. (For the SCCC, the free distance refers to the minimum Hamming weight of error events.) Thus the decrease in bit error probability for the SCCC is faster than the rate $1/N$, which represents a larger interleaver gain than that of the PCCC.

For decoding of the SCCC, a suboptimal decoding process based on iterations of the MAP algorithm can be applied to each constituent code similarly to the iterative decoder for the PCCC [26].

D. Bit Error Probability with SCCC

An upper bound on bit error probability for the SCCC can be found by modifying an analysis for serially concatenated block codes. For a convolutional code with code rate p/n and memory size m_c , there is an equivalent $(N/(p/n), N - pm_c)$ block code whose codewords are all sequences of length $N/(p/n)$ of the convolutional code starting from and terminating at the zero state [26]. The codewords of this equivalent block code are concatenations of error events of the convolutional code.

For N sufficiently larger than m_c , the number of codewords with input Hamming weight i (respectively k) and parity check bits of Hamming weight k (resp. d) for the outer code (resp., the inner code) is approximated by

$$A_o(i, k) \approx \sum_{x_0=1}^{x_0, \max} \binom{N/p}{x_0} A_{i, k, x_0}, \quad (31)$$

respectively,

$$A_i(i, k) \approx \sum_{x_i=1}^{x_i, \max} \binom{N/p}{x_i} A_{k, d, x_i} \quad (32)$$

where A_{i, k, x_0} (resp., A_{k, d, x_i}) is the number of sequences of weight k (resp. d), input weight i (resp. k), and the number of concatenated error events x_0 (resp. x_i) for the outer code (resp., the inner code). The ratio N/p appearing in (31) and (32) comes from the fact that N bits correspond to N/p input words for a code rate of p/n .

From this, the bit error probability for the SCCC can be upper bounded by

$$P_b \leq \sum_{d=0}^{N/R_i} \sum_{i=0}^{NR_0} \frac{i}{NR_0} A_1(i, d) P'_{b, u} \quad (33)$$

where $A_1(i, d)$ is the number of codewords generated by an input information word of Hamming weight i whose parity check bits have Hamming weight d , so that the overall Hamming weight is $i + d$, which is given by

$$A_1(i, d) = \sum_{k=d_f, o}^N \frac{A_0(i, k) A_i(k, d)}{\binom{N}{k}}. \quad (34)$$

VI. SIMULATION RESULTS

In this section, we present some simulation results that illustrate the performance of the systems described above. For these simulation examples, we use 8-ary PPM and OOCs with length $F = 500$ and, unless otherwise stated, weight $w = 5$. For 8-ary PPM, every three encoder bits are mapped to one of the eight ($= 2^3$) possible PPM symbols because the cardinality of the turbo encoder is set to be the same as that of the PPM modulator. The source bits are assumed to be independently interleaved and encoded by two constituent convolutional encoders. As discussed above, a rate-1/3 PCCC is employed, formed by two identical four-state recursive convolutional codes with generating polynomials $1 + D^2$ and $1 + D + D^2$. Figs. 4–10 show the upper bounds on turbo-coded bit error probabilities for the PCCC and the SCCC using (23) and (33), respectively.

In Fig. 4, the bit error probability is shown versus the average number of signal photons/nat ($K_s = K_s / \log_e M$) with $K = 10$ users, interleaver size $N = 100$, average number of noise photons $K_b = 50$, and for several different numbers of iterations used in the decoding process using the MAP algorithm. The uncoded case is also shown, and it is seen that turbo coding offers a considerable coding gain over an uncoded system. As the number of iterations in the decoding process increases, the performance is gradually improved through increased coding gain. However, it should be noted that, once the number of iterations exceeds some number (8, in this case), further iterations offer only marginal additional coding gain, as one would expect as the useful information is progressively gleaned from the observations.

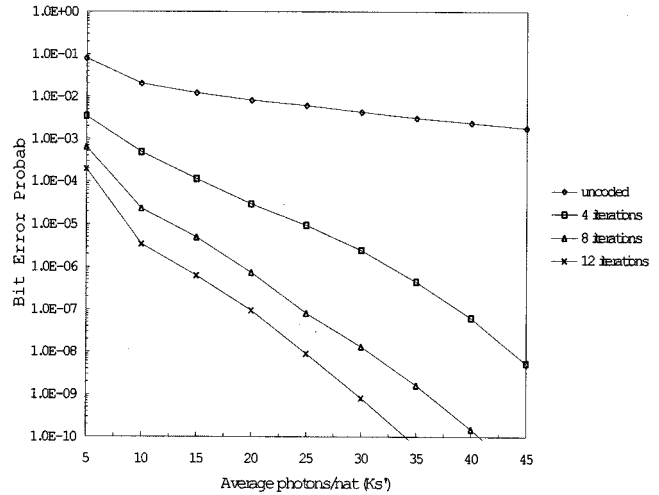


Fig. 4. Bit error probability versus average number of photons/nat for differing numbers of decoding iterations. ($K = 10$, $N = 100$, $K_b = 50$, $w = 5$, and using the MAP algorithm.)

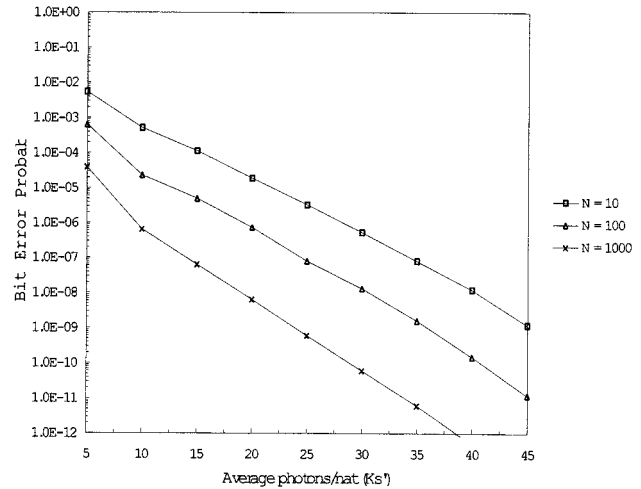


Fig. 5. Bit error probability versus average number of photons/nat for differing interleaver sizes. ($K = 10$, $K_b = 50$, $w = 5$, and eight decoding iterations using the MAP algorithm.)

In Fig. 5, the bit error probability is shown versus the average number of signal photons/nat, again with $K = 10$, $K_b = 50$, with eight decoding iterations using the MAP algorithm, and for differing interleaver sizes. These results confirm that, for a fixed code rate, performance is significantly improved by increasing the interleaver length. This result implies that significant performance improvement can be achieved by changing simple hardware structure, which is a major advantage of turbo codes.

In Fig. 6, the bit error probability is shown versus the average number of signal photons/nat, again with $K = 10$, $N = 100$, $K_b = 50$, eight iterations using the MAP algorithm, and for differing weights of the OOCs. It is seen that performance improves as the weight increases. This phenomenon stems from the fact that the average number of photons from the interfering signals decreases as the weight increases.

In Fig. 7, the bit error probability is shown versus the average number of signal photons/nat, again with $N = 100$, $K_b = 50$,

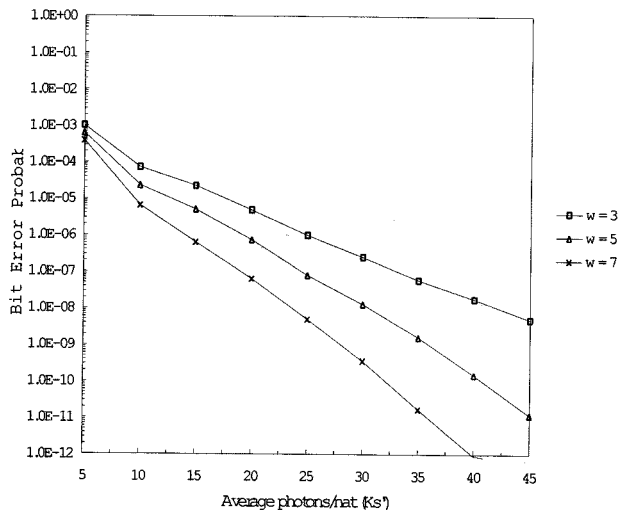


Fig. 6. Bit error probability versus average number of photons/nat for differing OOC weights. ($K = 10$, $N = 100$, $K_b = 50$, and eight decoding iterations using the MAP algorithm.)

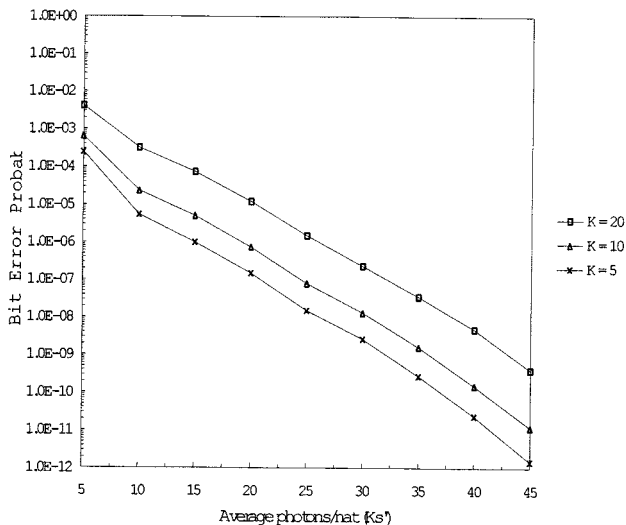


Fig. 7. Bit error probability versus average number of photons/nat for a differing numbers of users. ($N = 100$, $K_b = 50$, $w = 5$, and eight decoding iterations using the MAP algorithm.)

eight iterations using the MAP algorithm, and for differing numbers of users. (In this and subsequent figures, the weight of the OOCs is returned to $w = 5$.) These results show that increasing the multiple access interference has significant impact on the system performance, as one would expect.

In Fig. 8, the bit error probability versus average number of signal photons/nat is compared for differing average numbers of noise photons K_b , again with $K = 10$, $N = 100$, and eight iterations using the MAP algorithm. It is seen from these results that the performance deteriorates significantly as the number of background noise photons increases. For the case of very high noise photocounts such as $K_b = 200$, the coding gain from the turbo coding becomes very small because in this extreme case, the signal photocounts are almost merged into noise photocounts.

In Fig. 9, the bit error probability versus the average number of signal photons/nat is compared for different decoding al-

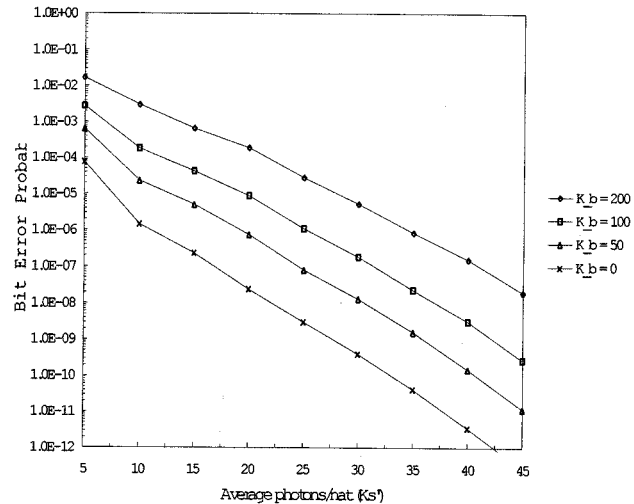


Fig. 8. Bit error probability versus average number of photons/nat for a differing average numbers of noise photons. ($K = 10$, $N = 100$, $w = 5$, and eight decoding iterations using the MAP algorithm.)

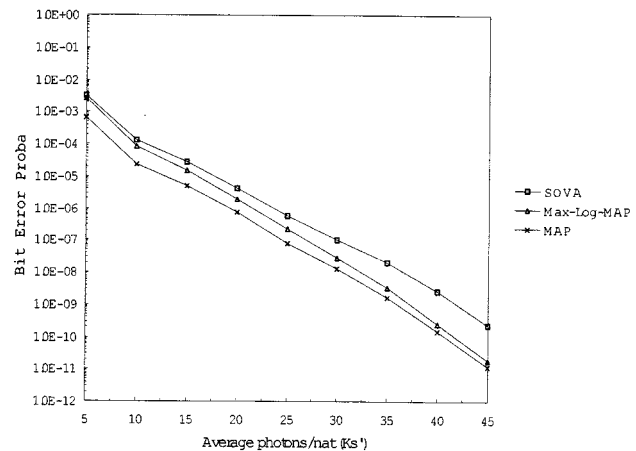


Fig. 9. Bit error probability versus average number of photons/nat for different decoding algorithms. ($K = 10$, $N = 100$, $K_b = 50$, $w = 5$, and eight decoding iterations.)

gorithms, again with $K = 10$, $N = 100$, $K_b = 50$, and eight iterations. It is seen from these results that, as expected, the MAP decoding algorithm achieves the best performance among the three algorithms; however, the performance difference between MAP and Max-Log-MAP becomes negligible as the number of signal photons increases. These results suggest that the Max-Log-MAP algorithm is a good overall choice when one considers both performance and complexity. The observed bit error probability difference between the MAP algorithm and the SOVA comes from the basic property, noted above, that the MAP algorithm minimizes bit error probability while the SOVA minimizes sequence error probability.

In Fig. 10, the bit error probability versus the average number of signal photons/nat is compared for serial and parallel concatenations of the constituent encoders, again with $K = 10$, $N = 100$, $K_b = 50$, and the MAP algorithm. The interleaver sizes of the SCCC and PCCC are assumed to be identical. The SCCC encoder consists of two identical four-state recursive convolutional codes with an inner encoder of code rate $2/3$ and an outer

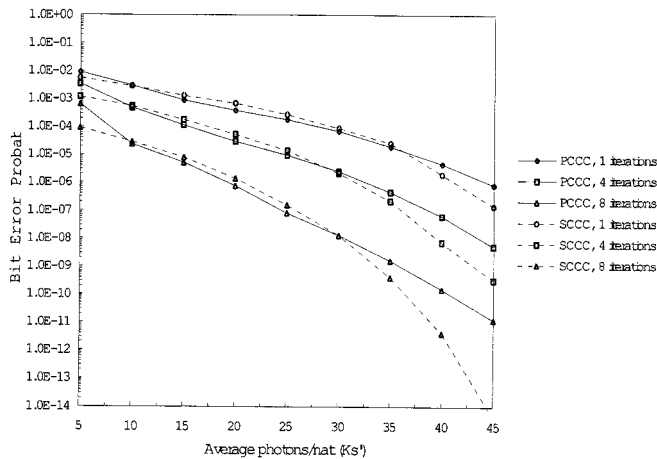


Fig. 10. Comparison of bit error probability between SCCC and PCCC. ($K = 10$, $N = 100$, $K_b = 50$, $w = 5$, and using the MAP algorithm.)

encoder of code rate $1/2$, while the PCCC encoder consists of two identical four-state convolutional codes with code rate $1/2$. Therefore, both the SCCC and PCCC have an overall code rate of $1/3$. For decoding of the SCCC, the decoding algorithm described in [25] is used. It is seen from Fig. 10 that the SCCC is comparable to the PCCC, and can be superior to it, especially when the signal photocounts are large. This phenomenon is due primarily to the higher interleaver gain for the SCCC.

VII. CONCLUSION

The performance of a turbo-coded optical CDMA system using PPM in the IM/DD optical channel has been analyzed and simulated. From the simulation results, we can conclude that turbo coding offers considerable coding gain with reasonable encoding and decoding complexity. From this coding gain, the required laser pulse power is reduced, which leads to the longer life expectancy for the laser system. Also, we have demonstrated that the bit error probability performance is substantially improved by increasing the interleaver length for a fixed code rate, which implies that a simple modification of the turbo encoder/decoder can be used to achieve the performance of a single convolutional code with much larger memory size (and therefore, much higher decoding complexity). We have also seen that the bit error probability performance is significantly enhanced as the number of iterations used in the decoding process increases, and that the Max-Log-MAP algorithm is a good overall choice for the constituent decoders. Finally, we have also seen that the SCCC is very competitive with the PCCC, and is superior to it when the signal photocounts are large.

The results of this paper can be extended in a number of ways, including encoding with more than two constituent encoders, or through the use of more sophisticated multiuser receiver techniques [27], [28]. However, the results can be interpreted generally as favoring the use of turbo codes in optical CDMA networks, and are applicable, for example, to optical wireless LANs. The authors have recently analyzed the throughput-delay characteristics of such an optical CDMA network using turbo coding in [29].

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